

Tax Distribution Pressure and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Distribution Pressure (TDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TDP achieves an annualized gross (net) Sharpe ratio of 0.37 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 39 (33) bps/month with a t-statistic of 3.65 (3.16), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, gross profits / total assets, Long-term EPS forecast, Change in equity to assets, Operating leverage, Asset growth) is 25 bps/month with a t-statistic of 2.27.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantaneously reflect all available information as predicted by the efficient market hypothesis. Instead, information diffuses gradually through markets, creating predictable patterns in asset returns as investors slowly incorporate new signals into prices. While the literature documents numerous instances of delayed price adjustment to earnings announcements, macroeconomic news, and firm fundamentals, we know relatively little about how tax-related events affect price formation. Tax distributions from mutual funds and ETFs create selling pressure that potentially contains information about underlying asset values, yet the speed at which markets incorporate these tax-induced trading flows remains unexplored. This gap presents an opportunity to examine whether tax distribution pressure represents a distinct source of return predictability arising from slow information diffusion.

We hypothesize that Tax Distribution Pressure (TDP) predicts future stock returns through gradual information incorporation mechanisms documented in the limited attention literature. When funds distribute capital gains to avoid tax penalties, they create selling pressure that temporarily depresses prices below fundamental values. Following the framework of [Hirshleifer and Teoh \(2003\)](#) on limited investor attention and [Hong et al. \(2000\)](#) on gradual information diffusion, we expect markets to underreact to these tax-driven flows initially, with prices adjusting slowly as information percolates through the investor base. The behavioral finance literature, particularly [Barber and Odean \(2008\)](#) on attention-driven trading and [DellaVigna and Pollet \(2009\)](#) on investor inattention, suggests that complex tax events receive less scrutiny than earnings releases or analyst recommendations, amplifying the underreaction. This slow diffusion mechanism predicts return continuation following high TDP events, as prices gradually incorporate the information content of tax-motivated trades. The effect should concentrate among stocks with greater

informational frictions—small firms, those with low analyst coverage, and limited institutional ownership—where [Hong and Stein \(1999\)](#) document that information travels most slowly. Furthermore, [Peng and Xiong \(2006\)](#) show that investors facing capacity constraints allocate attention strategically, often overlooking technical factors like tax distributions in favor of more salient fundamental news, creating persistent mispricing that sophisticated investors can exploit.

Our empirical analysis reveals that Tax Distribution Pressure strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on TDP achieves an annualized gross Sharpe ratio of 0.37, with the strategy earning average monthly returns of 36 basis points (t-statistic = 2.84). The predictive power remains robust after controlling for standard risk factors, with the strategy generating a monthly alpha of 39 basis points (t-statistic = 3.65) relative to the Fama-French five-factor model plus momentum. Notably, the TDP strategy maintains economic and statistical significance even after controlling for the six most closely related anomalies from the factor zoo, producing an alpha of 25 basis points per month (t-statistic = 2.27). The performance persists across different portfolio construction methodologies, with twenty-two out of twenty-five specifications yielding t-statistics exceeding two. Transaction cost analysis demonstrates that the strategy remains profitable after implementation costs, with net returns ranging from 9 to 46 basis points per month depending on portfolio construction. Among large-cap stocks—those above the 80th NYSE percentile—the TDP strategy delivers average returns of 23 basis points per month, confirming that the effect extends beyond small, illiquid securities where limits to arbitrage are most severe.

This paper makes three distinct contributions to the asset pricing and market efficiency literatures. First, we extend the gradual information diffusion framework of [Hong et al. \(2000\)](#) and [Hong et al. \(2007\)](#) by documenting a novel channel through which tax-induced trading flows create predictable return patterns, complementing

existing work in the Journal of Finance on institutional trading pressure and price formation. Our findings relate closely to [Coval and Stafford \(2007\)](#) in the Journal of Financial Economics, who document price pressure from mutual fund flows, but we identify a specific tax-motivated mechanism that generates more persistent return predictability. Second, we contribute to the growing literature on investor inattention pioneered by [DellaVigna and Pollet \(2009\)](#) in the Journal of Financial Economics and [Hirshleifer et al. \(2009\)](#) in the Review of Financial Studies by showing that complex tax events represent an overlooked source of mispricing that persists even in liquid markets. Unlike [Cohen et al. \(2003\)](#) in the Journal of Finance who focus on supply shocks from index additions, we document that tax distributions create both information signals and trading pressure that markets fail to immediately incorporate. Third, our comprehensive analysis using the [Novy-Marx and Velikov \(2023\)](#) protocol demonstrates that TDP represents a genuinely new anomaly that cannot be explained by existing factors or closely related predictors. The strategy’s robust performance net of transaction costs and its ability to enhance mean-variance efficient portfolios even when combined with 156 existing anomalies suggests that tax distribution pressure captures a distinct dimension of mispricing. These findings have important implications for market efficiency theories, suggesting that even well-understood mechanical trading events can generate exploitable return predictability when investor attention is limited.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of annual observations. The COMPUSTAT

database serves as the primary source for accounting and financial information used in constructing our Tax Distribution Pressure signal. The signal relies on two key financial statement variables. Income tax federal (TXFED) represents the federal income tax expense reported by firms in their financial statements for the fiscal year. This item captures the federal tax liability recognized under generally accepted accounting principles, reflecting the company’s tax obligations to the federal government. Dividends common (DVC) represents the total cash dividends declared on common stock during the fiscal year. This variable measures the cash distributions made to common shareholders, excluding any special or one-time dividends. We construct the Tax Distribution Pressure signal by dividing federal income tax expense (TXFED) by common dividends (DVC). This ratio captures the relationship between a firm’s federal tax obligations and its distributions to shareholders, providing insight into the relative burden of tax payments compared to dividend payouts. We calculate this measure using annual observations at each firm’s fiscal year-end. To ensure data quality, we require non-missing values for both TXFED and DVC, and we exclude observations where DVC equals zero to avoid undefined ratios. The resulting signal measures how many dollars of federal income tax expense a firm incurs per dollar of common dividends distributed, potentially reflecting constraints on the firm’s ability to maintain or increase dividend payments given its tax obligations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TDP signal. Panel A plots the time-series of the mean, median, and interquartile range for TDP. On average, the cross-sectional mean (median) TDP is 3.89 (0.95) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TDP data. The signal’s interquartile range spans -0.00 to 3.72. Panel B of Figure 1 plots the time-series of

the coverage of the TDP signal for the CRSP universe. On average, the TDP signal is available for 2.29% of CRSP names, which on average make up 5.31% of total market capitalization.

4 Does TDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TDP portfolio and sells the low TDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short TDP strategy earns an average return of 0.36% per month with a t-statistic of 2.84. The annualized Sharpe ratio of the strategy is 0.37. The alphas range from 0.25% to 0.41% per month and have t-statistics exceeding 2.06 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.59, with a t-statistic of -8.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 210 stocks and an average market capitalization of at least \$826 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 20 bps/month with a t-statistics of 2.39. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 9-46bps/month. The lowest return, (9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.10. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in

seventeen cases.

Table 3 provides direct tests for the role size plays in the TDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TDP, as well as average returns and alphas for long/short trading TDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TDP strategy achieves an average return of 23 bps/month with a t-statistic of 1.76. Among these large cap stocks, the alphas for the TDP strategy relative to the five most common factor models range from 11 to 34 bps/month with t-statistics between 0.87 and 2.94.

5 How does TDP perform relative to the zoo?

Figure 2 puts the performance of TDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TDP strategy falls in the distribution. The TDP strategy’s gross (net) Sharpe ratio of 0.37 (0.35) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TDP strategy (red line).² Ignoring trading costs, a \$1 invested in the TDP strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$6.79 which ranks the TDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TDP strategy would have yielded \$5.46 which ranks the TDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TDP relative to those. Panel A shows that the TDP strategy gross alphas fall between the 49 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases TDP ranks between the 73 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TDP with 201 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TDP or at least to weaken the power TDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TDP conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDP}TDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TDP. Stocks are finally grouped into five TDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDP trading strategies conditioned on each of the 201 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TDP signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

0.58, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on TDP is 2.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the TDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TDP strategy earns alphas that range from 29-40bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.75, which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TDP trading strategy achieves an alpha of 25bps/month with a t-statistic of 2.27.

7 Does TDP add relative to the whole zoo?

Finally, we can ask how much adding TDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes TDP grows to \$2314.57.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TDP is available.

8 Conclusion

This study provides robust evidence that Tax Distribution Pressure (TDP) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that TDP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on TDP delivers impressive performance metrics, with an annualized Sharpe ratio of 0.37 gross (0.35 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. The strategy’s ability to generate monthly abnormal returns of 39 basis points gross relative to the Fama-French five-factor model plus momentum, with a highly significant t-statistic of 3.65, underscores the signal’s robustness and statistical reliability. Particularly noteworthy is the persistence of alpha (25 bps monthly, t-statistic = 2.27) even after controlling for twelve factors, including six established risk factors and six closely related strategies from the factor zoo. This suggests that TDP captures unique information about expected returns not subsumed by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, TDP represents a novel addition to the growing literature on tax-related market inefficiencies and behavioral biases in equity markets. For practitioners, the signal’s robust performance and reasonable net returns after transaction costs suggest potential implementation value in quantitative investment strategies, particularly for investors seeking exposure to tax-driven market anomalies.

However, several limitations warrant consideration. First, while our analysis controls for transaction costs, real-world implementation may face additional frictions

such as market impact and capacity constraints that could erode returns. Second, the persistence of the TDP effect in out-of-sample periods and across different market regimes requires further investigation. Third, the underlying economic mechanisms driving the TDP anomaly deserve deeper theoretical and empirical exploration to better understand whether this represents a risk premium, behavioral bias, or institutional friction.

Future research should explore several promising avenues. Investigation into the time-varying nature of the TDP premium and its relationship with tax policy changes could provide valuable insights into the signal's economic drivers. Cross-country analysis would help determine whether TDP effects manifest similarly across different tax regimes and market structures. Additionally, examining the interaction between TDP and other tax-related anomalies could enhance our understanding of how tax considerations influence asset prices. Finally, research into optimal portfolio construction methods that incorporate TDP alongside other factors could yield practical insights for multi-factor investment strategies. As markets evolve and tax regulations change, continued monitoring of the TDP signal's efficacy will be essential to assess its long-term viability as an investment factor.

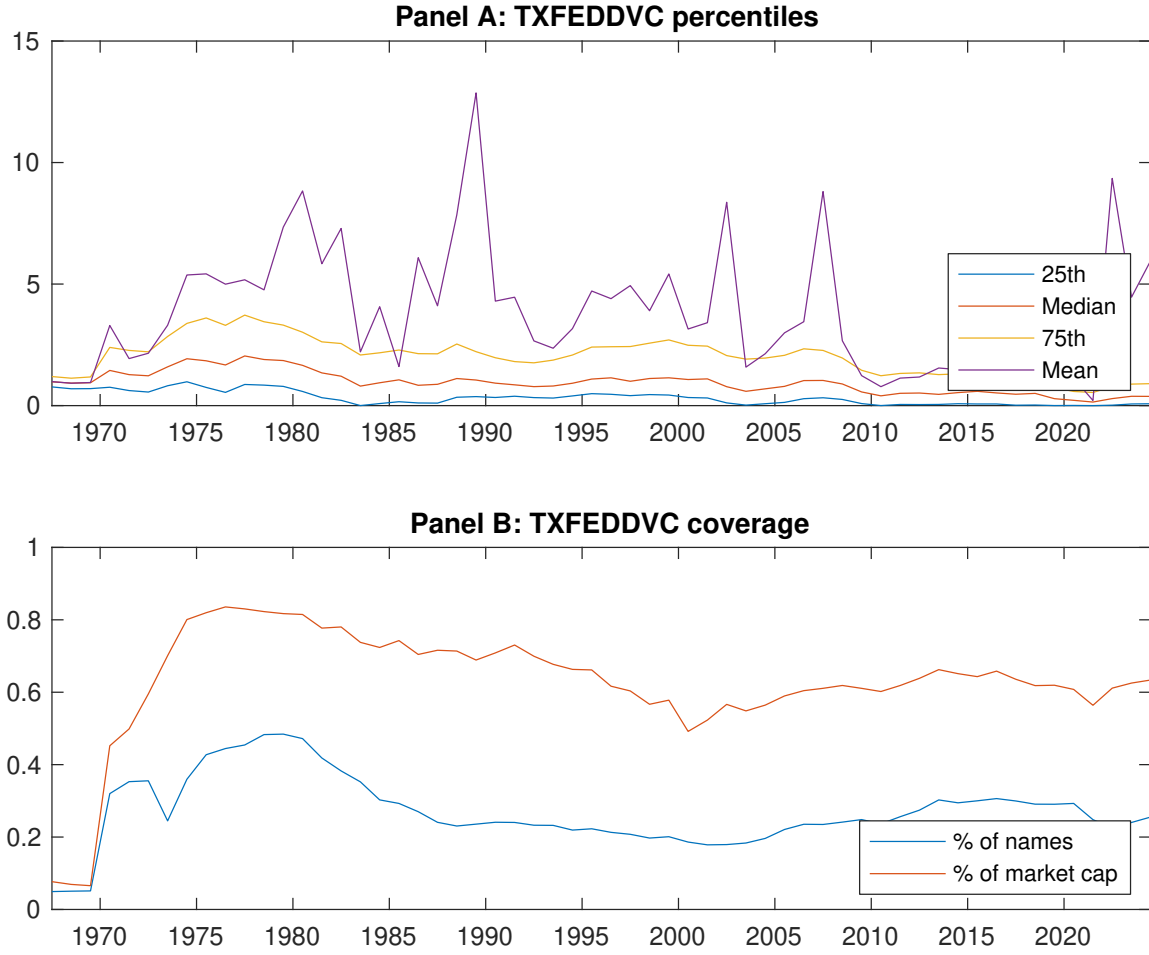


Figure 1: Times series of TDP percentiles and coverage.
This figure plots descriptive statistics for TDP. Panel A shows cross-sectional percentiles of TDP over the sample. Panel B plots the monthly coverage of TDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.61]	0.54 [3.44]	0.53 [3.22]	0.66 [3.63]	0.83 [3.99]	0.36 [2.84]
α_{CAPM}	-0.07 [-0.81]	0.07 [0.96]	0.04 [0.50]	0.09 [1.37]	0.18 [2.30]	0.25 [2.06]
α_{FF3}	-0.20 [-2.80]	0.01 [0.12]	-0.01 [-0.11]	0.09 [1.36]	0.20 [2.47]	0.40 [3.56]
α_{FF4}	-0.16 [-2.18]	-0.01 [-0.12]	-0.00 [-0.07]	0.12 [1.87]	0.24 [2.88]	0.40 [3.43]
α_{FF5}	-0.26 [-3.81]	-0.15 [-2.69]	-0.19 [-3.04]	-0.02 [-0.28]	0.14 [1.82]	0.41 [3.85]
α_{FF6}	-0.22 [-3.12]	-0.15 [-2.62]	-0.18 [-2.76]	0.02 [0.31]	0.18 [2.20]	0.39 [3.65]
Panel B: Fama and French (2018) 6-factor model loadings for TDP-sorted portfolios						
β_{MKT}	1.01 [62.05]	0.93 [67.81]	0.94 [61.84]	1.00 [68.39]	1.07 [56.82]	0.06 [2.32]
β_{SMB}	-0.04 [-1.55]	-0.19 [-9.42]	-0.14 [-6.13]	-0.00 [-0.10]	0.15 [5.35]	0.19 [4.99]
β_{HML}	0.15 [4.79]	0.04 [1.50]	0.02 [0.83]	-0.10 [-3.55]	-0.03 [-0.94]	-0.19 [-3.78]
β_{RMW}	-0.10 [-3.20]	0.23 [8.74]	0.38 [12.81]	0.21 [7.17]	0.25 [6.87]	0.36 [7.18]
β_{CMA}	0.48 [10.09]	0.35 [8.79]	0.25 [5.66]	0.18 [4.22]	-0.11 [-2.05]	-0.59 [-8.02]
β_{UMD}	-0.07 [-4.44]	-0.00 [-0.24]	-0.02 [-1.56]	-0.06 [-3.79]	-0.05 [-2.54]	0.02 [0.96]
Panel C: Average number of firms (n) and market capitalization (me)						
n	240	210	226	256	322	
me (\$10 ⁶)	826	1627	1752	1537	1107	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.36 [2.84]	0.25 [2.06]	0.40 [3.56]	0.40 [3.43]	0.41 [3.85]	0.39 [3.65]
Quintile	NYSE	EW	0.20 [2.39]	0.17 [2.09]	0.24 [3.04]	0.20 [2.43]	0.17 [2.20]	0.13 [1.68]
Quintile	Name	VW	0.37 [2.82]	0.25 [1.96]	0.40 [3.45]	0.40 [3.40]	0.42 [3.77]	0.41 [3.64]
Quintile	Cap	VW	0.26 [2.30]	0.16 [1.46]	0.30 [2.96]	0.29 [2.83]	0.34 [3.49]	0.32 [3.27]
Decile	NYSE	VW	0.50 [3.45]	0.41 [2.86]	0.60 [4.64]	0.64 [4.85]	0.60 [4.72]	0.63 [4.85]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.33 [2.62]	0.22 [1.81]	0.36 [3.16]	0.36 [3.13]	0.35 [3.28]	0.33 [3.16]
Quintile	NYSE	EW	0.09 [1.10]	0.07 [0.83]	0.13 [1.60]	0.11 [1.32]	0.05 [0.60]	0.03 [0.33]
Quintile	Name	VW	0.34 [2.60]	0.22 [1.71]	0.36 [3.04]	0.36 [3.07]	0.35 [3.19]	0.34 [3.11]
Quintile	Cap	VW	0.24 [2.09]	0.15 [1.31]	0.27 [2.68]	0.27 [2.66]	0.29 [3.01]	0.28 [2.88]
Decile	NYSE	VW	0.46 [3.19]	0.38 [2.61]	0.55 [4.20]	0.57 [4.38]	0.53 [4.17]	0.53 [4.24]

Table 3: Conditional sort on size and TDP

This table presents results for conditional double sorts on size and TDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TDP and short stocks with low TDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TDP Quintiles					TDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.82 [3.17]	0.90 [4.15]	0.83 [4.13]	1.26 [3.40]	0.93 [3.90]	0.11 [0.64]	0.08 [0.42]	0.12 [0.66]	0.08 [0.41]	0.12 [0.67]	0.08 [0.43]
	(2)	0.73 [3.30]	0.84 [4.23]	0.79 [3.90]	0.81 [3.80]	0.82 [3.51]	0.10 [0.80]	0.06 [0.47]	0.13 [1.08]	0.10 [0.85]	0.05 [0.39]	0.02 [0.20]
	(3)	0.68 [3.32]	0.65 [3.56]	0.77 [4.07]	0.93 [4.59]	0.83 [3.76]	0.15 [1.30]	0.10 [0.83]	0.19 [1.72]	0.19 [1.65]	0.09 [0.78]	0.09 [0.77]
	(4)	0.68 [3.41]	0.62 [3.52]	0.71 [3.87]	0.71 [3.71]	0.76 [3.50]	0.08 [0.63]	0.00 [0.02]	0.12 [0.96]	0.12 [1.01]	0.02 [0.19]	0.03 [0.25]
	(5)	0.47 [2.71]	0.52 [3.32]	0.49 [2.94]	0.61 [3.42]	0.70 [3.39]	0.23 [1.76]	0.11 [0.87]	0.26 [2.21]	0.26 [2.15]	0.34 [2.94]	0.32 [2.75]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TDP Quintiles					TDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	85	86	86	85	86	6	7	7	8	8	
	(2)	44	44	44	44	44	19	20	20	20	20	
	(3)	39	40	40	39	40	44	44	45	45	44	
	(4)	39	39	39	39	39	119	120	121	119	116	
(5)	43	43	43	43	43	973	1373	1191	1377	981		

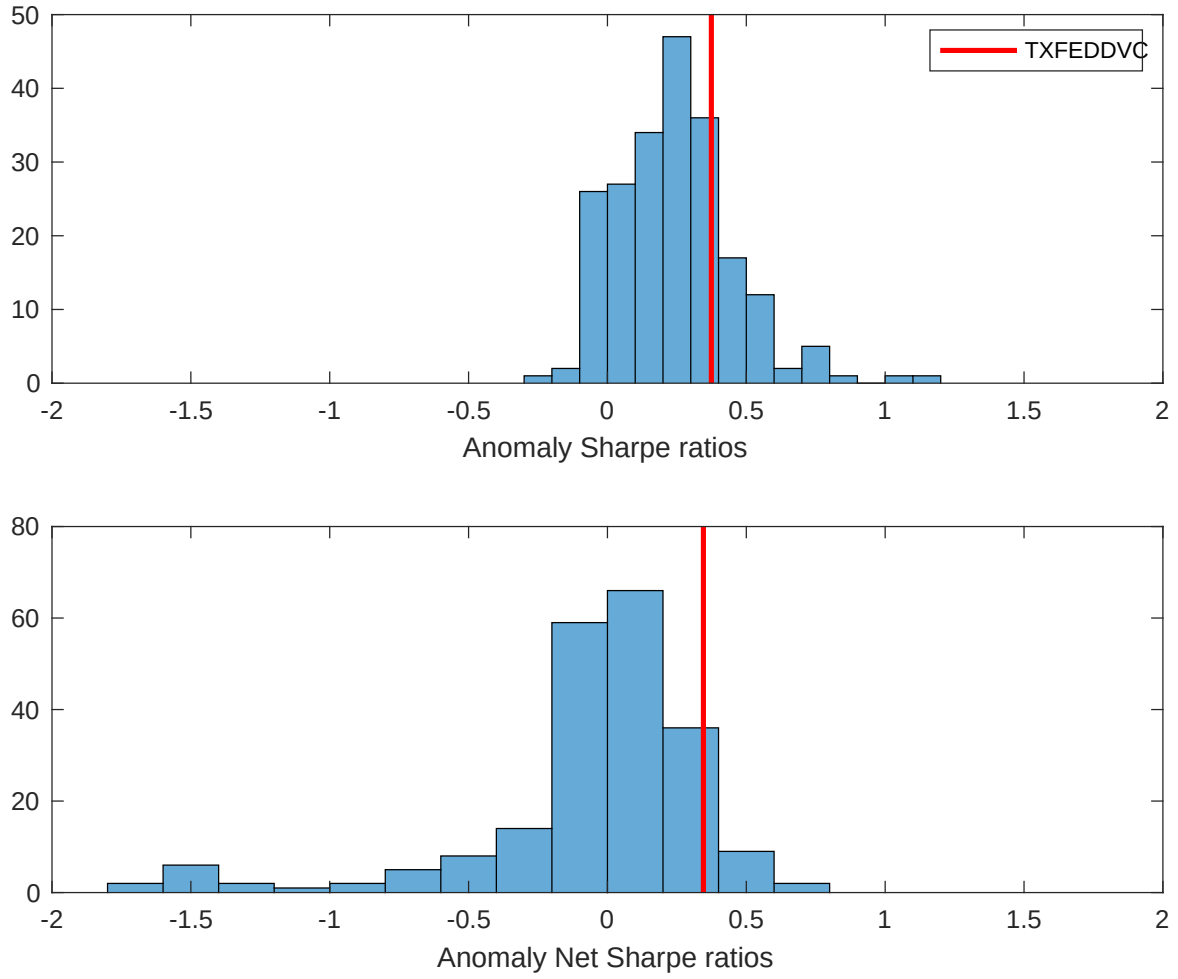


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

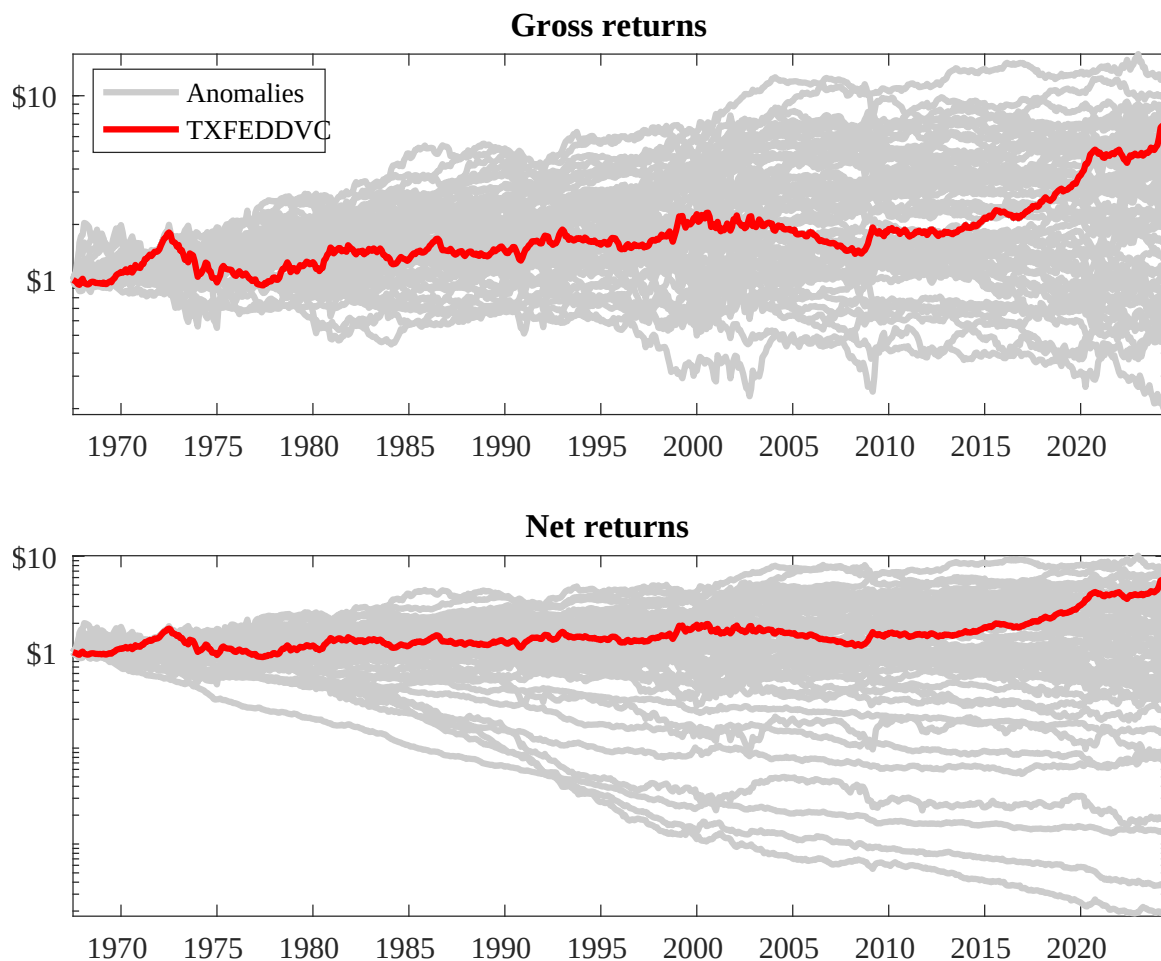
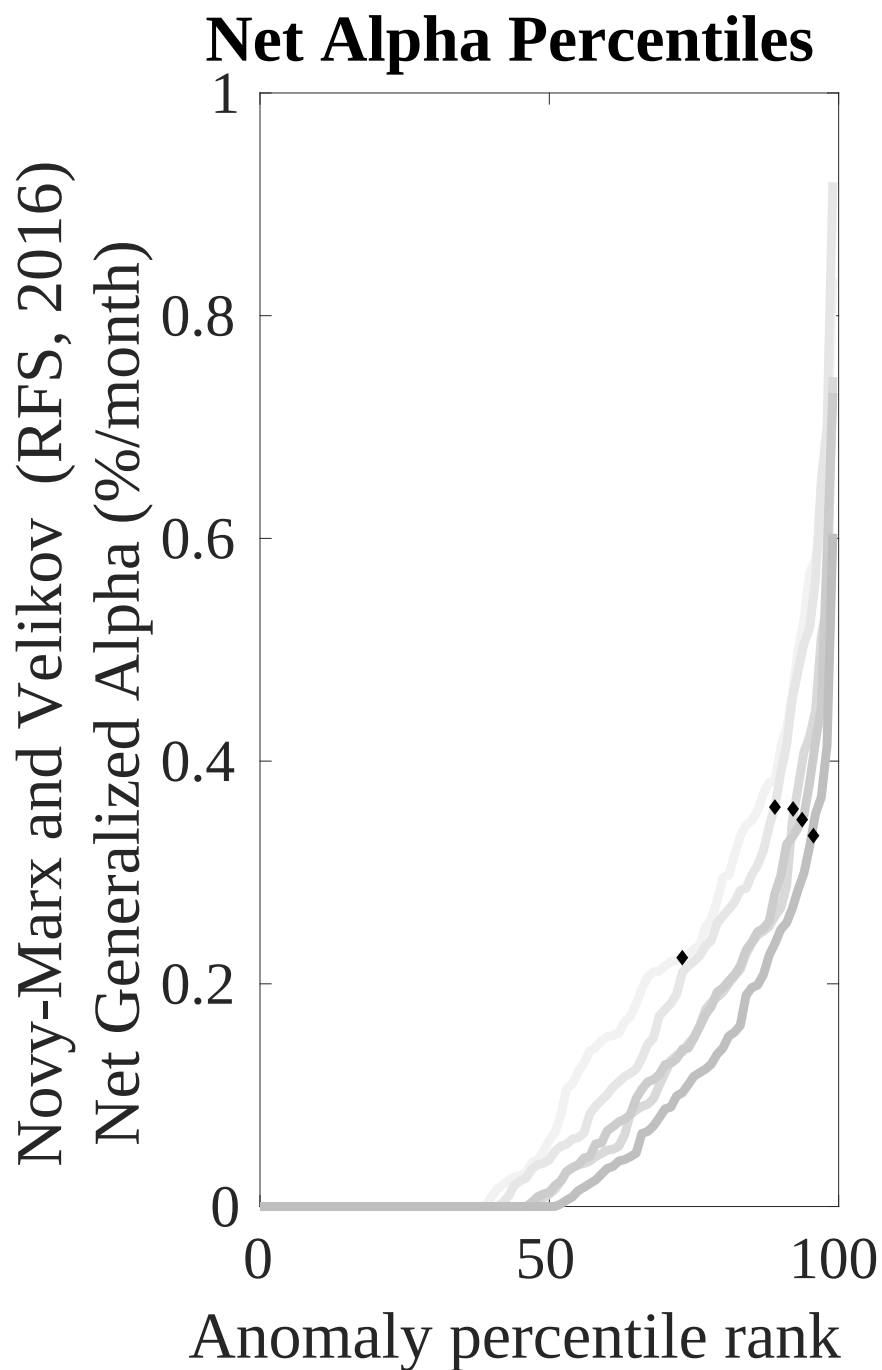
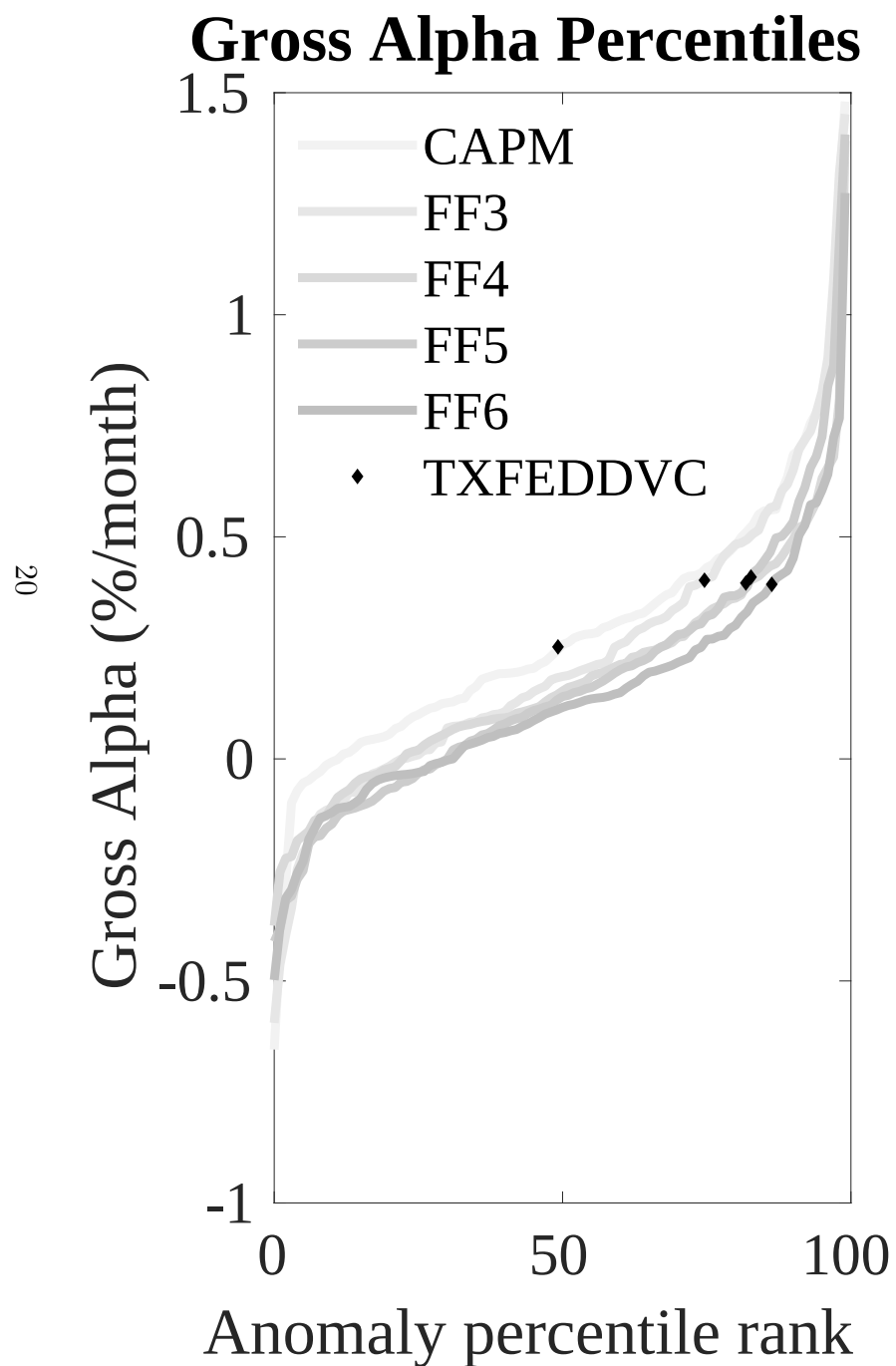
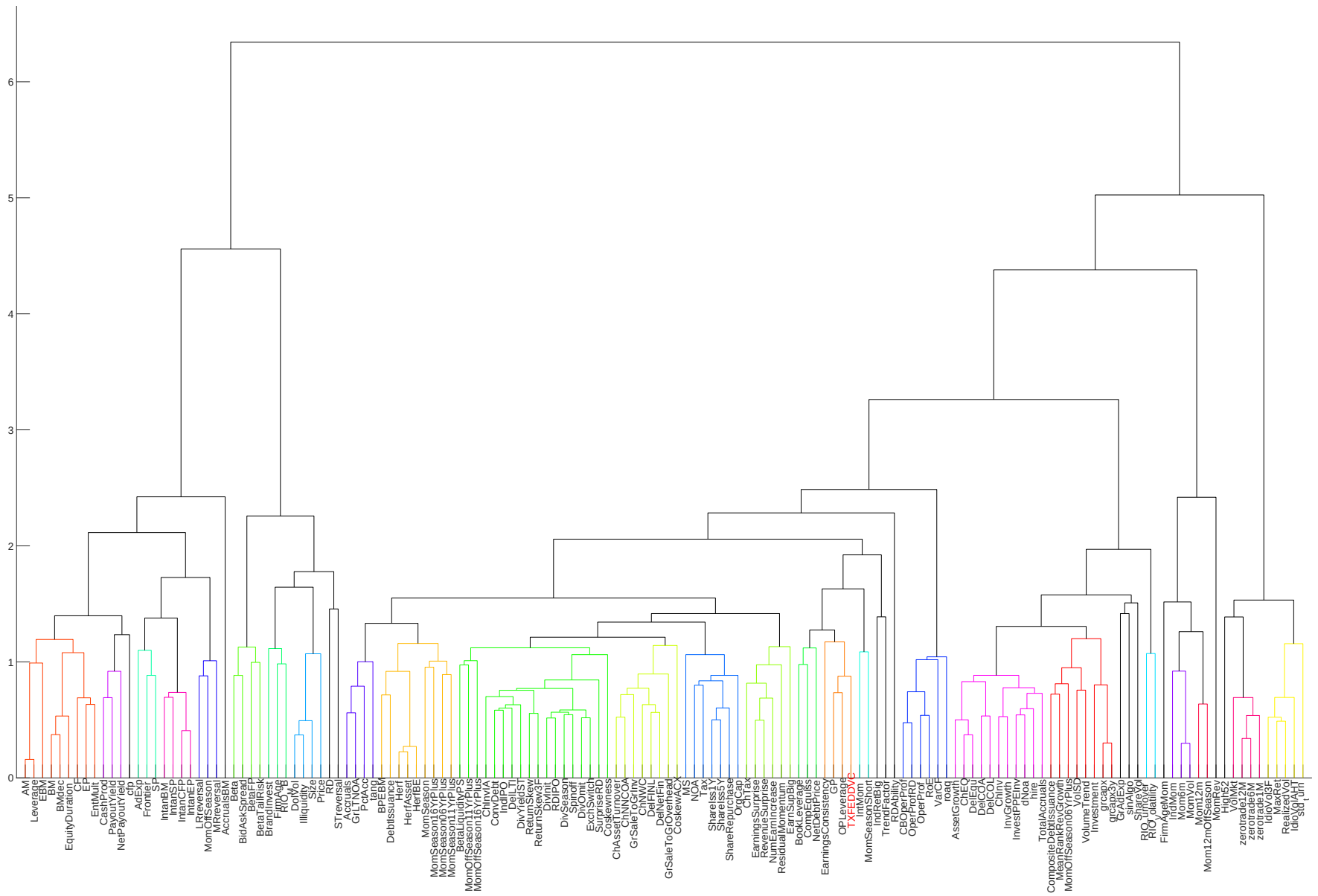


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TDF trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.





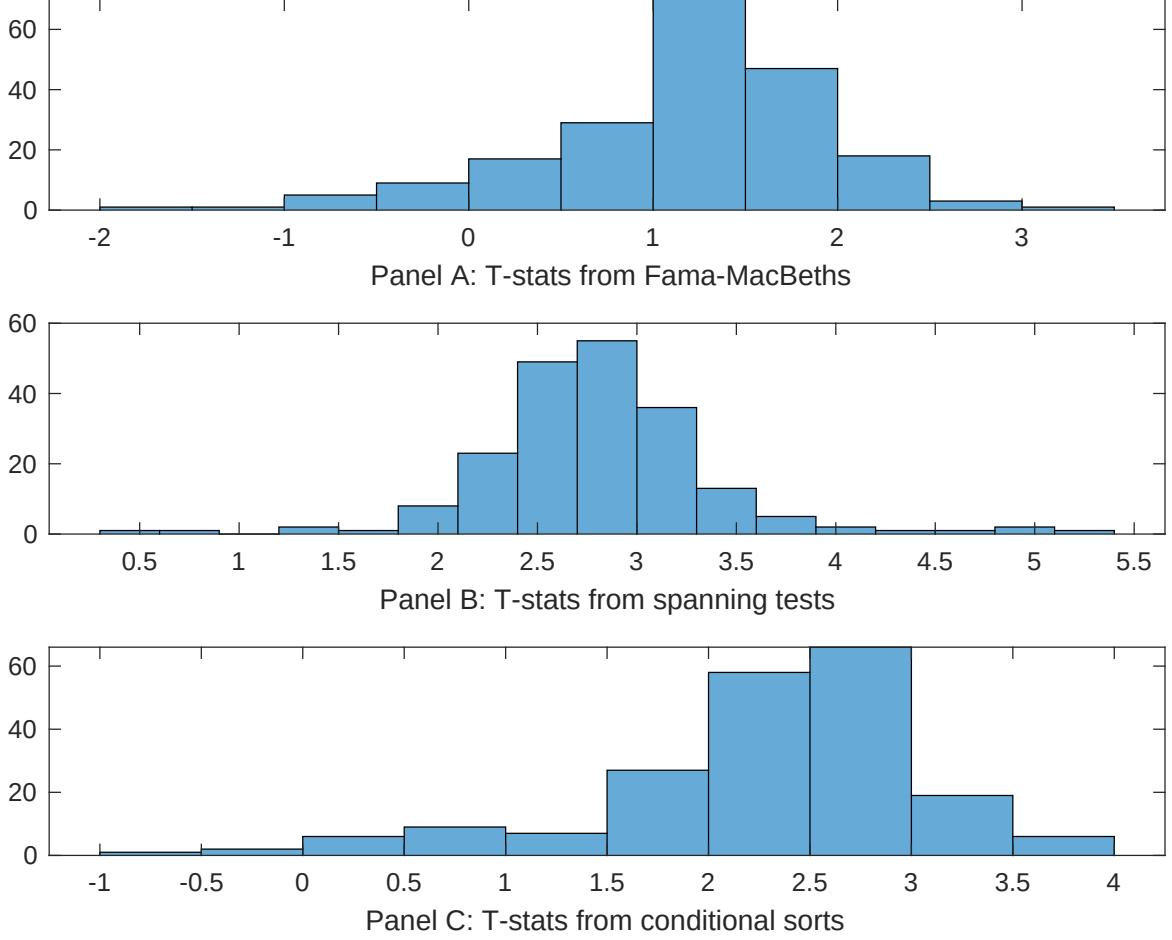


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TDP conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDP}TDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TDP. Stocks are finally grouped into five TDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDP trading strategies conditioned on each of the 201 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TDP, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TDP}TDP_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, gross profits / total assets, Long-term EPS forecast, Change in equity to assets, Operating leverage, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.18 [6.60]	0.10 [5.69]	0.13 [7.05]	0.12 [6.44]	0.11 [5.81]	0.12 [6.78]	0.12 [5.83]
TDP	0.34 [2.05]	0.12 [0.58]	0.35 [2.41]	0.37 [2.22]	0.22 [1.29]	0.32 [1.86]	0.36 [2.12]
Anomaly 1	0.59 [2.98]						0.37 [0.29]
Anomaly 2		0.42 [1.17]					0.22 [1.29]
Anomaly 3			1.00 [1.31]				0.64 [0.83]
Anomaly 4				0.16 [2.53]			0.46 [0.86]
Anomaly 5					0.16 [1.76]		0.22 [0.44]
Anomaly 6						0.84 [5.91]	0.29 [2.26]
# months	684	684	504	684	684	684	504
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, gross profits / total assets, Long-term EPS forecast, Change in equity to assets, Operating leverage, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.40 [3.78]	0.29 [2.75]	0.34 [2.90]	0.39 [3.65]	0.35 [3.39]	0.39 [3.66]	0.25 [2.27]
Anomaly 1	-22.66 [-3.84]						-51.25 [-5.22]
Anomaly 2		33.97 [8.09]					25.74 [4.64]
Anomaly 3			-31.37 [-7.87]				-20.50 [-4.72]
Anomaly 4				-2.95 [-0.55]			45.91 [4.63]
Anomaly 5					31.54 [7.55]		20.24 [4.07]
Anomaly 6						-0.84 [-0.13]	15.49 [2.03]
mkt	5.15 [2.04]	5.30 [2.19]	-6.24 [-2.04]	5.94 [2.34]	6.29 [2.58]	5.89 [2.32]	-4.31 [-1.48]
smb	19.74 [5.33]	17.64 [4.93]	-1.83 [-0.42]	19.15 [5.12]	5.56 [1.39]	19.19 [5.08]	-5.69 [-1.29]
hml	-15.78 [-3.20]	-2.54 [-0.50]	4.42 [0.79]	-17.95 [-3.60]	-0.31 [-0.06]	-18.30 [-3.70]	21.71 [3.75]
rmw	35.04 [7.11]	16.55 [3.11]	34.77 [6.36]	35.86 [7.18]	18.55 [3.49]	36.16 [7.27]	9.73 [1.51]
cma	-37.04 [-3.98]	-55.72 [-7.92]	-44.59 [-5.33]	-56.40 [-6.14]	-66.97 [-9.39]	-58.41 [-5.49]	-71.33 [-6.68]
umd	2.74 [1.09]	1.80 [0.74]	9.02 [3.36]	2.23 [0.88]	2.13 [0.88]	2.28 [0.90]	8.08 [3.19]
# months	684	684	504	684	684	684	504
$\bar{R}^2(\%)$	36	40	40	35	40	34	48

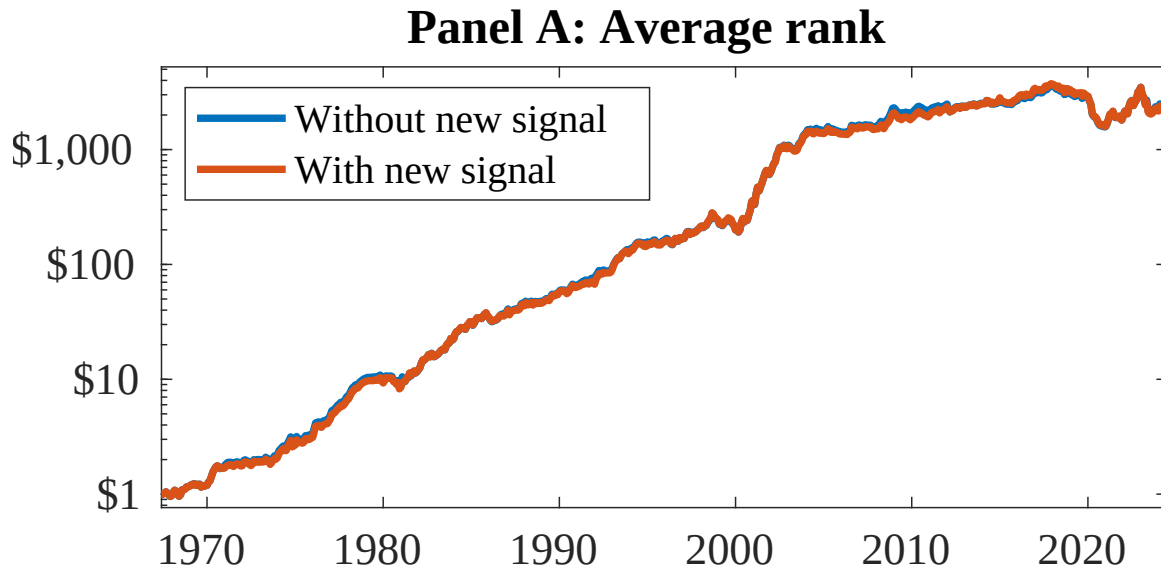


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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